

LeetCode 130: Surrounded Regions

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Short Notes & Logic

- **Core Logic:** The approach is similar to the "Closed Islands" problem. Start from the boundaries because any 'O' connected to the boundary cannot be captured.
- **Step 1 (Boundary DFS):** Traverse the borders of the matrix. If an 'O' is found, run a DFS to mark it and all its connected (*saath me juda hua*) 'O's as Safe ('S').
- **Step 2 (Identify Unsafe Regions):** At the end of the boundary traversals, we will have some or no 'O's left in the inner grid which are not safe (meaning they are completely surrounded by 'X'). We need to convert them.
- **Step 3 (Final Conversion):** Iterate through the entire board one last time:
 - Convert any remaining unprotected 'O' to 'X' (*jo nahi bach paaye convert ho gye*).
 - Convert the marked safe 'S' cells back to 'O' (*jinpe shield thi wo O reh gaye*).

Source Code

```
class Solution {
public:

    vector<pair<int,int>> dir={{1,0},{-1,0},{0,1},{0,-1}};
    int n;
    int m;

    void dfs(int r, int c, vector<vector<char>>& board) {
        board[r][c] = 'S'; //marked them as safe
        for(auto it : dir) {

            int nr = r + it.first;
            int nc = c + it.second;
            if(nr >= 0 && nc >= 0 && nr < n && nc < m) {

                if(board[nr][nc] == 'O') {
                    dfs(nr, nc, board);
                }
            }
        }
    }

    void solve(vector<vector<char>>& board) {
        //i am gonna use the same approach as closed islands
        //i.e start from boundary and any 'O' can make its saath me juda hua 'O' a
        //at the end we will have some or no 'O' which are not safe , we need to

        n = board.size();
        m = board[0].size();

        for(int i=0; i<n; i++) {
            if(board[i][0] == 'O') {
                dfs(i, 0, board);
            }

            if(board[i][m-1] == 'O') {
                dfs(i, m-1, board);
            }
        }

        for(int j=0; j<m; j++) {
```

```

        if(board[0][j] == 'O') {
            dfs(0, j, board);
        }

        if(board[n-1][j] == 'O') {
            dfs(n-1, j, board);
        }

    }

    for(int i=0; i<n; i++) {
        for(int j=0; j<m; j++) {
            if(board[i][j] == 'O') {
                board[i][j] = 'X'; //jo nahi bach paaye convert ho gye
            }

            else if(board[i][j] == 'S') {
                board[i][j] = 'O'; //jinpe shield thi wo O reh gaye
            }
        }
    }

}

};

```